

# **ENV 6106**

**Air Pollution Modeling**

**Spring 2023**

# Topics

- AERMOD
- Expectation
  - Understand the most important pathway and keywords
  - Be able to run AERMOD at least for a simple case
- Reference:
  - AERMOD user guide
  - AERMOD quick reference guide
  - AERMOD implementation guide

# Components of AERMOD

- AERMET: meteorological data preprocessor
  - AERSURFACE: for surface characteristics
  - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
  - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

# Input data to AERMOD

- Meteorology
  - Surface layer parameters & upper air profile
  - Commonly named *SFC* and *PFL* files
  - Generated by AERMET
- Terrain
  - Elevation for emission sources and receptors
    - Also hill height scale for receptors
  - Generated by AERMAP
- Building parameters
  - Used for estimating building downwash
  - Generated by BPIPPRIM

# Input data to AERMOD

- Receptor locations
  - Where concentrations will be estimated
  - Can be gridded or discrete, Cartesian or polar
- Source information
  - Source types: point, area, volume, line, open pit
    - Point source can be POINT, POINTCAP, POINTHOR
    - Area source can be AREA, AREAPOLY, AREACIRC
    - Line source can be LINE, BUOYLINE (inherited from BLP model in 2015)
  - Source ID, location, height and other characteristics
  - Emission rates and temporal variations
    - Adjust by a combinations of factors, or specified hourly rates

# AERMOD control file

- AERMOD uses similar control files as in AERMAP
- No interactive interface, reads “aermod.inp” file
  - AKA run stream input file
- The control file contains pathways and keywords
- Pathways are only two letters
  - CO: overall job **C**ontrol
  - SO: **S**ource information
  - RE: **R**Eceptor information
  - ME: **M**Eteorology
  - OU: **O**utput information
  - EV: **E**Vent processing (for source contributions)
    - We will not go over event processing in this class

**CO pathway**

# AERMOD CO pathway

- Overall job control
- Start and end CO pathway
  - CO STARTING
  - CO FINISHED
- Between start and finish
  - CO TITLEONE *a bunch of title info*
    - Information to identify this AERMAP run
  - CO TITLETWO *more title info*
    - More information if needed, optional

# AERMOD CO pathway

- CO MODELOPT: model options
  - DFAULT: Use regulatory modeling options
    - Options not consistent with regulatory will be ignored
  - CONC: calculate concentration
  - DEPOS: calculate total deposition flux
  - DDEP/WDEP: calculate dry or wet deposition flux
  - FLAT/ELEV: whether elevated terrain is used
  - BETA: use options that are fancy but still under test
  - SCREEN: run AERMOD in screen mode
    - Essentially run AERSCREEN in a more complicated way

# AERMOD CO pathway

- CO AVERTIME: averaging time
  - Can be 1, 2, 3, 4, 6, 8, 12, 24
    - Corresponding to 1,2.....24 hour average
  - Can also be MONTH, ANNUAL or PERIOD
    - Monthly, annual average, or ave across entire run period
  - Different ave time may be required for different pol
  - NAAQS standards for CAPs:
    - <https://www.epa.gov/criteria-air-pollutants/naaqs-table>
- CO RUNORNOT
  - Do a setup check before model execution?
  - Do check: CO RUNORNOT NOT
  - No check: CO RUNORNOT RUN

# AERMOD CO pathway

- CO POLLUTID: name of pollutants
  - Use certain names triggers certain ave algorithm
  - PM<sub>2.5</sub>: use PM25, PM2.5, PM-2.5, or PM-25
    - 24-h std: 98th percentile, averaged over 3 years
  - NO<sub>2</sub>: use NO2, then OLM or PVMRM can be used
    - 1-h std: 98th percentile of 1-hour daily maximum concentrations, averaged over 3 years
  - SO<sub>2</sub>: SO2, 4-hour half-life time usually used
    - 1-h std: 99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
- CO ERRORFIL: list detailed messages

# Example CO pathway

CO STARTING

TITLEONE A simple and quick aermoc test case

MODELOPT CONC FLAT DFAULT

AVERTIME 1 PERIOD

POLLUTID CO

RUNORNOT RUN

ERRORFIL UCF\_TEST\_ERRORS.TXT

CO FINISHED

**SO pathway**

# AERMOD SO pathway

- Specify emission source information
- Start and end SO pathway
  - SO STARTING
  - SO FINISHED
- Source location for non-line sources:
  - SO LOCATION SrcID Srctyp Xs Ys (Zs)
  - SrcID: name of source
  - Srctyp: source type (point, line, area etc)
  - Xs, Ys, Zs: coordinates of source, elevation (Z) is optional
- Source location for line sources:
  - SO LOCATION SrcID Srctyp Xs1 Ys1 Xs2 Ys2 (Zs)
  - Xs1&2, Ys1&2: coordinates of two ends of line source

# AERMOD SO pathway

- Point source emission information
  - SO SRCPARAM
  - Point: SRCPARAM Srcid Ptemis Stkhgt Stktmp Stkvel Stkdia
  - *Ptemis*: emission rate in grams / second
  - *Stkhgt, Stktmp, Stkvel, Stkdia*: stack height, exit gas temperature and velocity, and inside stack diameter
- Emission information for other source types also uses SRCPARAM, with different formats
  - See AERMOD user guide section 3.3.2
- Adjustment factors can be specified to adjust emission rates temporally

# AERMOD SO pathway

- Building information
  - Copy and paste from BPIPPRM output
- Include external source file
  - SO INCLUDED *filename*
- Grouping sources
  - SO SRCGROUP *SrcGrpID SrcID's SrcRange's*
  - *SrcGrpID*: name of source group
  - *SrcID's, SrcRange's*: individual and/or range of source ID
    - E.g. SRC1 SRC2 SRC3 = SRC1-SRC3
  - Can use ALL to combine all sources into one group
  - SRCGROUP has to be last keyword before SO FINISHED

# Example SO pathway

SO STARTING

LOCATION FAKESRC POINT 480415 3163945

\*\* Point Source                      QS      HS      TS      VS      DS

\*\* Parameters:                      ----      ----      ----      ----      ---

SRCPARAM FAKESRC 100.0   65.0   350.   10.0   1

SRCGROUP ALL

SO FINISHED

**RE pathway**

# AERMOD RE pathway

- Specify receptor information
- Start and end RE pathway
  - RE STARTING
  - RE FINISHED
- Can copy and paste from AERMAP output
- Receptor specified in the same way as AERMAP

Recall these two examples

```
RE GRIDCART NetID STA
                XYINC Xinit Xnum Xdelta Yinit Ynum Ydelta
END
```

```
RE GRIDCART NetID STA
                XPNTS Gridx1 Gridx2 ..... Gridxm
                YPNTS Gridy1 Gridy2 ..... Gridyn
END
```

# Example RE pathway

RE STARTING

GRIDCART REC1 STA

XYINC 480015 10 40 3163545 10 40

END

RE FINISHED

**ME pathway**

# AERMOD ME pathway

- Specify meteorology input data
- Start and end ME pathway
  - ME STARTING
  - ME FINISHED
- Surface met data:
  - ME SURFFILE: specify surface met data file
  - ME SURFDATA: surface met station information
    - SURFDATA StationID Year (Name) (Xcoord Ycoord)
    - Recommend station ID & year match what's in *SURFFILE*
    - Name and X, Y are not used, for identification

# AERMOD ME pathway

- Upper air met data:
  - ME PROFFILE: specify upper air met data file
  - ME UAIRDATA: upper air station information
    - UAIRDATA StationID Year (Name) (Xcoord Ycoord)
- Others
  - ME PROFBASE
    - Base elevation above MSL for calculation potential temperature profile
    - Should set to elevation of main met station
  - ME STARTEND: start and end processing time
    - STARTEND Strtyr Strtmn Strtdy (Strthr) Endyr Endmn Enddy (Endhr)

# Example ME pathway

ME STARTING

SURFFILE MCO2017\_V16216.SFC

PROFFILE MCO2017\_V16216.PFL

SURFDATA 12815 2017 MCO

UAIRDATA 12842 2017 TAMPA

PROFBASE 0 METERS

ME FINISHED

**OU pathway**

# AERMOD OU pathway

- Specify output information, all keywords are optional
  - But at least have some output, or AERMOD stops
- Start and end OU pathway
  - OU STARTING
  - OU FINISHED
- Highest value summary by receptor
  - OU RECTABLE Aveper nth.....
  - *Aveper*: which short term averaging time period needed. Use ALLAVE when all specified time period under CO pathway
  - *nth.....*: which rank of highest value wanted. Can specify as (1,2,3...) or (FIRST,SECOND,THIRD....) or (1ST,2ND,3RD,4TH...) or a range (1-3)

# AERMOD OU pathway

- Highest value summary by receptor
  - OU RECTABLE Aveper nth.....
- Combined with CO POLLUTID, the nth keyword is useful for NAAQS compliance modeling for some pollutants
  - PM<sub>2.5</sub>: use PM25, PM2.5, PM-2.5, or PM-25
    - 24-h std: 98th percentile, averaged over 3 years
  - NO<sub>2</sub>: use NO2, then OLM or PVMRM can be used
    - 1-h std: 98th percentile of 1-hour daily maximum concentrations, averaged over 3 years
  - SO<sub>2</sub>: use SO2, 4-hour half-life time can be used
    - 1-h std: 99th percentile of 1-hour daily maximum concentrations, averaged over 3 years

*See section 3.2.15 & 3.2.16 of AERMOD user guide for more information*

*Be aware, AERMOD changes, always refer to the latest AERMOD user guide*

# AERMOD OU pathway

- Overall highest value summary
  - OU MAXTABLE: Aveper Maxnum
  - *Maxnum*: output top nth maximum values
- Output raw results at each receptor for post-processing
  - OU POSTFILE Aveper Grpid Format Filnam
  - *Grpid*: source group ID
  - *Format*: UNFORM for unformatted, or PLOT for formatted
  - *Filnam*: name of output file

# Example OU pathway

OU STARTING

RECTABLE ALLAVE 1-3

MAXTABLE ALLAVE 20

POSTFILE 1 ALL PLOT RAWDAT.DAT

OU FINISHED

# Final remarks

- We went over most of the AERMOD system
  - Still some left, such as LEADPOST
- Rarely someone know all AERMOD system inside and out when just started doing modeling
- And model keeps evolving!
- Fear not, federal & local EPA has many helpful staff and guidance documents available